

A Comprehensive Review of Deep Learning-Based Forest Fire and Smoke Detection: Model Architectures, Hardware Backends, and Cross-Platform Benchmarking

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ABSTRACT

Wildfires rank among the most ecologically and economically destructive natural disasters worldwide, fueling an acute demand for accurate real-time automated detection systems. This review offers an in-depth survey of the development of deep learning-based forest fire and smoke detection by synthesizing results across 6 seminal studies as well as our original experimental work comparing Metal Performance Shaders (MPS), CUDA and CPU hardware backends. We review the evolution from traditional sensor-based and colour-feature approaches, through shallow machine learning classifiers, to modern Convolutional Neural Networks (CNNs), attention mechanisms and hybrid spatio-temporal architectures. Transfer learning based on learning without forgetting (LwF). Compact and efficient CNN design for real-time deployment. Spectral pattern learning from satellite images using the Input Amplification (InAmp) module. Video based Detection and Temporal Accumulation (DTA) using Faster R-CNN and Long Short-Term Memory (LSTM). In our hardware benchmarking study, comparing Apple MPS (M4 Pro) vs. NVIDIA CUDA (RTX 3050 Notebook) and Apple CPU, across a custom ForestFireCNN on roughly 42,000 images, we show that MPS takes 12 minutes to complete training, while CUDA takes 48 minutes. We also show that all backends converge to roughly 96.5% validation accuracy, conclusively establishing hardware-agnostic model quality. The key research gaps identified in this review include the lack of quantitative energy profiling, limited cross-architecture benchmarking, and the underexplored intersection of spectral pattern learning and hardware-efficient deployment. Future work recommendations include multi-generational Apple Silicon benchmarking, integration of InAmp-style spectral preprocessing with hardware-optimised CNN pipelines and real-time edge deployment on MPS-equipped devices for environmental surveillance.

Keywords: *Forest fire detection, deep learning, convolutional neural networks, transfer learning, MPS, CUDA, hardware benchmarking, smoke detection, real-time detection, Apple Silicon, satellite imagery, spatiotemporal features.*

1. Introduction

Forest fires are among the most destructive of natural calamities on Earth, causing severe ecological disruption, loss of biodiversity, extensive economic damage and increasing contributions to atmospheric greenhouse gas concentrations. The frequency and intensity of wildfire events are increasing due to climate change, making early and accurate detection a critical global concern. Automated vision-based fire detection systems provide an attractive solution that can be deployed for continuous-scale surveillance, with response times an order of magnitude faster than traditional sensor-based alarms that rely on physical proximity to an active fire.

In the last ten years, a paradigm shift has been witnessed in the methodology of fire and smoke detection, with the arrival of deep learning and especially Convolutional Neural Networks (CNNs). Earlier systems relied on hand-crafted colour features, temporal pixel analysis and rule-based classifiers which are all highly sensitive to environmental variability such as lighting, shadow, and fire-like objects, while CNN-based approaches learn discriminative feature hierarchies directly from image data and provide better generalisation and accuracy. But as models become more complex, the demand for efficient computational infrastructure increases.

This review bridges three active research streams: (1) deep learning model architecture for fire and smoke detection, (2) hardware acceleration and cross-platform benchmarking for deep learning training, and (3) efficient deployment of vision systems in real-world surveillance environments. In our own experimental study, which is published in the International Research Journal of Modernization in Engineering Technology and Science (IRJMETS, November 2025), we present the first direct, controlled comparison of Apple Metal Performance Shaders (MPS), NVIDIA CUDA and CPU backends for a CNN based forest fire detection task using an identical model, dataset and hyperparameter configuration across all the three platforms.

The rest of this review is organized as follows. Section 2 describes the historical evolution of fire detection from traditional methods to deep learning. Section 3 reviews the CNN architectures and transfer learning strategies used for fire and smoke detection. In Section 4, we discuss efficient and compact CNN designs for real-time deployment. 5. Approaches Based on Satellite and Spectral Patterns. Section 6 reviews the spatiotemporal video-based detection methods. The results of our cross-platform benchmarking study and hardware acceleration are presented in Section 7. Section 8 provides a comparative synthesis across all the works considered. Section 9 discusses research gaps and future directions. Section 10 ends.

2. Evolution of Fire Detection: From Sensor Systems to Deep Learning

2.1.Traditional and Sensor-Based Approaches

Fire detection systems have been based in the past on three general categories: manual surveillance, physical sensor networks, and remote sensing using satellites. Sensor-based systems (thermal, smoke, gas or temperature detectors) provide high specificity in controlled environments but are inherently limited in detection range and subject to false alarms from non-fire stimuli (e.g., fog, steam and dust). Importantly they are not effective in open outdoor environments where forests are most at risk, and require spatial density to provide sufficient coverage.

Satellite systems such as MODIS and VIIRS have provided the means for large-scale fire surveillance on a global basis. However, as described in Kenchgunde et al. (2025), these systems suffer from limited temporal resolution, strong dependence on weather, and intrinsic delay in response times that degrade their utility in identifying fires in early stages. The early pixel-level fire smoke detection from satellite imagery used manually derived spectral-band threshold values, as detailed by Zhao et al. (2024), who point out that thresholds were handcrafted based on domain knowledge and experience, display significant variation across sensors and environmental conditions, and are thus fundamentally non-generalizable.

Computer vision techniques emerged as a viable alternative, making use of the properties of colour, shape and motion of fire and smoke. Kim and Lee (2019) reviewed early methods using RGB, HSI, YUV, YCbCr and CIElab colour spaces for segmenting candidate fire pixels. These colour-based approaches, although computationally efficient, are inherently sensitive to brightness variations, shadows and fire-like objects, especially in sunset scenarios, red-coloured objects and architectural lighting, resulting in unacceptably high false alarm rates in real-world outdoor deployments.

2.2.Shallow Machine Learning Methods

The limitations of purely rule-based colour methods motivated the adoption of shallow machine learning classifiers. Support Vector Machines (SVMs) were explored for separating fire candidate pixels from background regions using extracted colour-texture feature descriptors. Decision tree architectures provided non-temporal classification with 80-90% mean true positive detection but 7-8% false positive rates, as reported in Thomson et al. (2020). Neural networks with a single hidden layer were applied to pixel-level spectral classification in satellite imagery, demonstrating the importance of spectral patterns while being constrained by the absence of spatial context extraction.

A key observation from this period, synthesised across multiple reviewed studies, is that machine learning-based fire detection algorithms relied on manually extracted visible information from images, focusing only on shallow characteristics of flame that could lead to data loss during manual extraction. This motivated the transition to deep learning architectures capable of automatic hierarchical feature learning.

2.3.The Deep Learning Paradigm

The introduction of large-scale datasets and GPU-accelerated training enabled researchers to develop

deep CNNs that substantially outperformed all prior approaches. AlexNet, VGG16, ResNet50, and GoogLeNet architectures were among the first deep models applied to fire detection, establishing the viability of end-to-end deep feature learning for this task. Muhammad et al. (2018) proposed a GoogleNet-inspired fine-tuned CNN for fire detection in CCTV surveillance, demonstrating that transfer learning from ImageNet could elevate fire detection accuracy to 94.43% on benchmark datasets — a significant improvement over hand-crafted feature methods. Sathishkumar et al. (2023) further demonstrated that CNN-based models significantly outperform traditional feature-engineering approaches, achieving accuracies above 90% on diverse fire and smoke datasets using VGG16, InceptionV3, and Xception architectures with transfer learning.

3. CNN Architectures and Transfer Learning for Fire and Smoke Detection

3.1. Pre-trained Model Adaptation

The application of transfer learning to fire and smoke detection has been extensively studied by Sathishkumar et al. (2023), who systematically evaluated VGG16, InceptionV3, and Xception architectures as both feature extractors and fine-tuners on a custom 6,911-image fire and smoke dataset. Their work establishes several important benchmarks. In feature extractor mode, pre-trained weights are frozen and only a custom classification head is trained, yielding Xception validation accuracy of 98.27% and testing accuracy of 97.77%. In fine-tuner mode, the top 5, 8, and 7 layers of VGG16, InceptionV3, and Xception respectively are unfrozen and retrained, producing further improvements, with Xception achieving the highest testing accuracy of 98.72%.

The superiority of Xception in this context is attributed to its use of depth-wise separable convolutions, which decompose standard convolutions into depth-wise and point-wise operations, achieving greater representational capacity with fewer parameters than comparable architectures. The depth-wise separable convolution structure also facilitates faster and more accurate feature learning — a property that is directly relevant to the compact architecture design explored in Thomson et al. (2020).

A critical architectural comparison emerging from Sathishkumar et al. (2023) is presented in Table 1 below, summarising validation and testing accuracy across models and configurations.

Table 1: Comparative performance of transfer learning configurations on fire/smoke dataset (Sathishkumar et al., 2023)

Model	Mode	Validation Accuracy (%)	Testing Accuracy (%)
VGG16	Feature Extractor	95.18	94.38
VGG16	Fine-Tuner	96.32	95.46
InceptionV3	Feature Extractor	93.93	92.04
InceptionV3	Fine-Tuner	97.87	97.01
Xception	Feature Extractor	98.27	97.77
Xception	Fine-Tuner	99.12	98.72

3.2. Bayesian Hyperparameters Optimisation

A distinctive methodological contribution of Sathishkumar et al. (2023) is the application of Bayesian Optimisation for hyperparameter tuning — selecting learning rate, optimizer, activation function, batch size, number of epochs, and number of neurons in custom layers. Bayesian Optimisation was preferred over random search and grid search due to its ability to leverage information from previous model evaluations when selecting hyperparameter values for subsequent runs, enabling faster convergence to optimal configurations. The optimised hyperparameters — Adam optimizer, ReLU activation, and varying learning rates (1e-3 for Xception, 1e-5 for InceptionV3, 1e-1 for VGG16) — contributed substantially to the state-of-the-art results obtained, a contribution the authors note was absent from prior comparable studies.

3.3. Learning Without Forgetting

Perhaps the most theoretically significant contribution of Sathishkumar et al. (2023) is the integration

of Learning Without Forgetting (LwF) for transfer across datasets. LwF addresses the well-documented catastrophic forgetting phenomenon in deep learning — the tendency of a network to lose performance on its original training task when fine-tuned for a new related task. LwF accomplishes this by training the network using only new task data while simultaneously preserving task-specific parameters from the original dataset, optimising a combined loss function across both old and new tasks.

Applied to the challenging 240-image BoWFire dataset — which contains difficult fire-like scenarios including sunset conditions, fire-colored objects, and architectural lighting — Xception with LwF achieved 91.41% accuracy, compared to 79.23% without LwF. Crucially, original task accuracy on the forest fire dataset was preserved at 96.89%, demonstrating that LwF successfully maintains classification capability across both tasks simultaneously. This result has direct practical implications: a fire detection model trained on one domain can adapt to new deployment environments without catastrophic accuracy regression on the original task.

4. Compact and Efficient CNN Architectures for Real-Time Fire Detection

4.1. Motivation for Architecture Compression

While deep and accurate models like Xception achieve state-of-the-art classification performance, their deployment in real-world surveillance contexts demands a trade-off between accuracy and computational efficiency. Security camera systems, UAV-mounted sensors, and edge computing devices impose stringent constraints on memory, processing speed, and power consumption. Muhammad et al. (2018) explicitly motivated their GoogleNet-based architecture by the need for smaller model size and FPGA suitability. Thomson et al. (2020) extended this line of research by proposing two reduced-complexity CNN architectures specifically optimised for non-temporal real-time fire detection: NasNet-A-OnFire and ShuffleNetV2-OnFire.

4.2. NasNet-A-OnFire and ShuffleNetV2-OnFire

Thomson et al. (2020) applied transfer learning from ImageNet-pretrained NasNet-A-Mobile and ShuffleNetV2 architectures, experimentally simplifying each through filter pruning guided by L2-norm magnitude analysis. ShuffleNetV2-OnFire — the most efficient architecture proposed — achieved full-frame binary classification accuracy of 95% with only 155,617 total parameters, representing a 437-fold reduction compared to the FireNet baseline of 68.3 million parameters, while operating at 40 fps on CPU and 69 fps on GPU (NVIDIA 2080Ti). This represents an 8x improvement in computational efficiency over InceptionV1-OnFire. For superpixel-based in-frame fire localisation, ShuffleNetV2-OnFire achieved 94.4% accuracy at 18 fps GPU throughput.

Crucially, deployment on the low-powered NVIDIA Xavier-NX embedded device — with TensorRT FP16 quantisation — demonstrated 49 fps throughput for ShuffleNetV2-OnFire, validating the architecture for real-world edge deployment scenarios. This result directly demonstrates that compact CNN architectures designed for fire detection can operate at inference speeds adequate for real-time video surveillance without sacrificing meaningful accuracy, making them highly relevant for integration with the MPS-based on-device training pipelines demonstrated in Kenchunde et al. (2025).

Table 2: Computational efficiency of fire detection CNN architectures for full-frame binary classification (Thomson et al., 2020)

Architecture	Parameters (M)	Accuracy (%)	A:C Ratio	CPU fps	GPU fps
FireNet	68.3	91.5	1.3	17	-
InceptionV1-OnFire	1.2	93.4	77.9	8.4	-
NasNet-A-OnFire	3.2	95.3	29.78	7	35
ShuffleNetV2-OnFire	0.156	95.0	608.97	40	69

4.3.The GoogleNet-Inspired Surveillance CNN

Muhammad et al. (2018) proposed a cost-effective CNN architecture for fire detection in CCTV surveillance videos, motivated by the operational constraints of real-world surveillance networks. Their model — inspired by the GoogleNet/Inception architecture — consists of 100 layers with 2 main convolutions, 4 max pooling layers, one average pooling layer, and 7 inception modules. Transfer learning from a pre-trained GoogleNet model with fine-tuning over 10 epochs elevated detection accuracy from an initial 88.41% to 94.43% on Dataset1 (Foggia et al., 31 surveillance videos), simultaneously reducing the false alarm rate from 11% to 0.054% and false negatives from 5.5% to 1.5%.

The study demonstrated the robustness of the fine-tuned CNN against adversarial challenges including noise, cropping, rotation, and fire-colored objects — successfully detecting small amounts of fire placed on otherwise normal scenes and correctly classifying sunset images as non-fire. This resilience to false positives is particularly important in practical deployment, as it directly reduces unnecessary emergency dispatch — a concern also addressed by the temporal integration methods reviewed in Section 6.

5. Satellite-Based Fire Smoke Detection and Spectral Pattern Learning

5.1.The Spectral Challenge in Satellite Fire Smoke Detection

Fire smoke detection from satellite imagery presents a uniquely challenging classification problem. As documented by Zhao et al. (2024), fire smoke shares similar spatial and spectral characteristics with other aerosols — cloud, fog, haze, and dust — and frequently intermingles with these aerosols in satellite imagery, making accurate discrimination extremely difficult from spatial features alone. The shapes and colours of fire smoke vary dramatically across geographic regions, seasons, and fire types, indicating that spatial patterns alone are fundamentally insufficient for reliable satellite-based fire smoke detection.

Early pixel-level detection relied on manually derived spectral-band threshold values — handcrafted from domain knowledge to exploit distinctive spectral signatures of smoke versus confounding aerosols in specific satellite bands. While effective in the specific sensor and environmental contexts for which they were designed, these thresholds generalise poorly across different satellite sensors (MODIS, VIIRS, Landsat, Himawari-8) and environmental conditions. Neural network applications at the pixel level demonstrated that automatically extracted spectral features could achieve good accuracy, but early implementations lacked spatial context and required manually engineered input features.

5.2.The InAmp Module: Automated Class-Specific Spectral Pattern Learning

Zhao et al. (2024) proposed the Input Amplification (InAmp) module — a novel deep learning preprocessing block specifically designed to enable DL models to automatically learn class-specific pixel-level spectral patterns from multi-spectral satellite imagery. Unlike spatial feature extraction performed by conventional 3x3 CNN filters, InAmp operates through three successive attention mechanisms:

- Band attention: 1x1 convolution filters applied directly to individual pixels across spectral bands, extracting raw spectral patterns as learnable deep-pseudo bands without relying on predefined formulas or domain knowledge.
- Spatial attention: Refinement of extracted spectral patterns by learning the importance of each pixel within each spectral or deep-pseudo band based on spatial location, enhancing class-specific pixels while suppressing background.
- Channel attention: Weighting of the combined spectral patterns according to their significance for classification, guiding the model toward the most discriminative spectral combinations.

The InAmp module outputs 32 channels (determined through ablation studies as optimal across both evaluated datasets) and is designed to integrate seamlessly with existing CNN architectures as a lightweight preprocessing block, adding minimal additional parameters while providing substantial accuracy gains.

5.3.Experimental Evaluation Across Datasets and Architectures

The effectiveness of InAmp was validated on two fire smoke satellite imagery datasets:

USTC_SmokeRS (6,225 MODIS RGB images across 6 classes) and Landsat_Smk (1,836 Landsat 5/8 multi-spectral images with 6 spectral bands including RGB, NIR, SWIR-1, and SWIR-2). Four baseline CNN architectures were evaluated with and without InAmp: ResNet50, InceptionResNetV2, MobileNetV2, and the domain-specific lightweight model VIB_SD. For the Landsat_smk dataset — which offers the richer spectral information context most relevant to InAmp's spectral pattern extraction — all four models demonstrated improved accuracy and kappa coefficient when augmented with InAmp, with VIB_SD achieving the largest improvement from 81.79% to 85.33% accuracy and a reduction in the Smoke class false negative rate from 24.26% to 13.97%.

The authors note that fire smoke's distinct reflection characteristics in infrared bands — particularly NIR and SWIR — provide the spectral discriminability that InAmp learns to exploit. This has important implications for deployment in multi-spectral satellite platforms and highlights a research direction with direct relevance to the hardware-efficient on-device deployment discussed in Kenchgunde et al. (2025): InAmp-augmented lightweight models offer a path toward accurate spectral fire smoke detection on portable, power-efficient devices.

6. Video-Based and Spatiotemporal Fire Detection Methods

6.1.Limitations of Static Frame-Based Detection

A fundamental limitation shared by still-image CNN classifiers — including the architectures reviewed in Sections 3 and 4 — is their inability to capture the dynamic temporal behavior of fire and smoke. False positives arising from fire-like objects such as sunset conditions, chimney smoke, red-colored objects, and steam require temporal context for accurate discrimination: a fire's flickering motion, expanding spatial extent, and evolving smoke characteristics over time are definitive indicators that no single frame can fully capture. Kim and Lee (2019) explicitly motivate their video-based approach by observing that almost all prior research only uses a still image or consecutive pairs of frames to detect fire, considering only short-term dynamic behavior, whereas fire has a longer-term dynamic behavior critical to reliable discrimination from false positives.

6.2.Detection and Temporal Accumulation (DTA)

Kim and Lee (2019) proposed a deep learning-based fire detection method they term Detection and Temporal Accumulation (DTA), which explicitly models the human process of fire detection: identifying a suspected fire object, continuously monitoring it, and accumulating temporal observations before reaching a definitive conclusion. The DTA pipeline operates in three integrated stages.

In the first stage, Faster R-CNN — equipped with ResNet-101 as the convolutional backbone — detects and localises bounding boxes for three object classes: Flame, Smoke, and Non-fire (including difficult negative examples such as chimney smoke, sunset, and cloud). The Faster R-CNN was trained on 73,887 still images and achieved a mean Average Precision (mAP) of 88.3% across all three classes.

In the second stage, spatial features extracted via weighted Global Average Pooling (GAP) within the detected bounding boxes are sequentially fed to a two-stage LSTM network over a short temporal window (TLSTM = approximately 60 frames, or 1.5 seconds at 30fps). The LSTM captures short-term dynamic fire behavior and produces a binary fire/non-fire decision. Testing across 8,527 positive and 7,547 negative video clips demonstrated 95.00% accuracy with 512 LSTM hidden cells.

In the third stage, majority voting combines short-term LSTM decisions across a longer temporal window ($T_{\text{vot}} = 10$ seconds) for the final fire determination. This ensemble over time — directly analogous to a human's accumulation of multiple glances before concluding fire is present — substantially improves performance. Evaluated on the Foggia et al. benchmark dataset of 31 surveillance videos, the DTA method with majority voting achieved 97.92% accuracy with a false positive rate of 2.47% and false negative rate of 1.38%, outperforming all compared methods including the CNN-based approach of Muhammad et al.

Table 3: Performance comparison of video-based fire detection methods on Foggia et al. benchmark dataset (Kim and Lee, 2019)

Method	False Positive (%)	False Negative (%)	Accuracy (%)
Proposed DTA (LSTM-512)	3.04	1.73	95.00
Proposed DTA (Majority Voting, 10s)	2.47	1.38	97.92
Muhammad et al. CNN	8.87	2.12	94.50
Foggia et al. Multi-expert	11.67	0.00	93.55
Habibugle et al.	5.88	14.29	90.32

6.3. Dynamic Fire Behavior Interpretation

Beyond binary fire classification, the DTA framework provides dynamic fire behavior interpretation by separately computing weighted bounding box areas for Flame and Smoke objects across frames and reporting temporal changes in these areas. This capability — tracking whether flame area is increasing or decreasing over time — provides actionable information for emergency response: a detected fire that is diminishing in flame area but expanding in smoke area suggests active suppression, whereas increasing flame area signals escalation. Across 38 additional video clips with diverse fire dynamics, the majority voting mechanism correctly rejected false positive detections (clouds, steam, sunset, chimney smoke) consistently, demonstrating the reliability of temporal integration for complex real-world scenarios.

7. Hardware Acceleration and Cross-Platform Benchmarking for Deep Learning

7.1. Background: The Hardware Landscape for Deep Learning

The computational demands of training deep CNNs have historically been dominated by NVIDIA CUDA-enabled GPUs, whose massively parallel architecture and mature software ecosystem — including PyTorch and TensorFlow — have made them the de facto standard for deep learning research and production. CUDA enables thousands of simultaneous floating-point operations and has been central to the breakthroughs in deep learning accuracy documented across all models reviewed in this paper. However, CUDA's dominance has not gone unchallenged. Apple's Metal Performance Shaders (MPS) backend, introduced for the M-series Apple Silicon chips, provides GPU-based deep learning acceleration through the Metal API — with a fundamentally different memory architecture that eliminates the data transfer bottleneck between separate CPU and GPU memory pools inherent in discrete GPU designs.

The Apple M4 Pro chip — used in the experimental study of Kenchgunde et al. (2025) — integrates a 12-core CPU, 16-core GPU, and 24 GB unified memory within a single system-on-chip architecture. This unified memory model allows both CPU and GPU operations to access the same physical memory without explicit data transfer overhead, providing a potential efficiency advantage particularly relevant for moderate-scale deep learning workloads where data transfer latency is a meaningful component of total training time.

7.2. The ForestFireCNN Architecture

The model at the centre of the hardware benchmarking study — ForestFireCNN — is a compact custom CNN designed to balance cross-platform portability with adequate representational capacity for the three-class fire detection task. The architecture consists of a feature extraction block comprising three sequential convolutional layers (16, 32, and 64 filters with 3x3 kernels and padding of 1), each followed by ReLU activation and MaxPooling with kernel size 2, progressively reducing spatial dimensions from 128x128 to 16x16 while learning low-level (edges, colour intensity), mid-level (smoke texture, flame boundaries), and high-level (discriminative class features) representations. The classification block consists of a fully connected layer (16,384 to 128 neurons with ReLU), a Dropout layer (p=0.3) for regularisation, and a Softmax output layer for three-class probability prediction.

The intentional compactness of ForestFireCNN serves the benchmarking objective: by avoiding architecture-induced training bottlenecks, observed differences in training time across hardware backends can be more reliably attributed to hardware-level computational efficiency rather than confounded by model

complexity effects.

7.3.Dataset and Experimental Configuration

A comprehensive dataset of approximately 42,000 labeled images — distributed across Fire, Smoke, and Non-Fire classes — was assembled from multiple open-source repositories including the Mendeley Forest Fire Dataset. The training set comprised 32,398 images and the validation set 10,500 images. All images were resized to 128x128 pixels and normalised to [0,1] pixel intensity range. Data augmentation included random horizontal and vertical flips, rotations ($\pm 15^\circ$), brightness and contrast adjustments, and random cropping and zooming. All experiments used identical hyperparameters: 8 training epochs, batch size of 32, learning rate of 0.001, and Adam optimizer with CrossEntropyLoss, ensuring that observed performance differences are attributable to hardware backend rather than training configuration. Execution times were automatically logged using Python's time module.

7.4.Hardware Backend Performance Results

The experimental results across all three hardware backends are summarised in Table 4. The most striking finding is the 4-fold training time advantage of MPS over CUDA under the tested hardware conditions.

Table 4: Cross-platform benchmarking results for ForestFireCNN (Kenchgunde et al., 2025)

Hardware Platform	Backend	Avg. Epoch Time (s)	Total Training Time (min)	Validation Accuracy (%)	Final Val Loss
Apple M4 Pro (GPU)	MPS	~90	~12	96.56	0.099
Apple M4 Pro (CPU)	CPU	~151	~20	96.52	0.120
NVIDIA RTX 3050 (Notebook)	CUDA	~358	~48	96.39	0.104

The MPS backend completed training in approximately 12 minutes, compared to approximately 20 minutes for the M4 Pro CPU and 48 minutes for the NVIDIA RTX 3050 CUDA backend — a ratio of 1:1.67:4 respectively. Despite this four-fold difference in training speed, all three backends converged to near-identical validation accuracy: 96.56% (MPS), 96.52% (CPU), and 96.39% (CUDA). Final validation loss followed a similar pattern: 0.099 (MPS), 0.120 (CPU), 0.104 (CUDA).

The training and validation loss curves across all 8 epochs demonstrated consistent downward convergence for all three backends, confirming effective learning regardless of hardware. MPS achieved the fastest loss reduction and lowest final training loss (~0.10). Notably, MPS reached its accuracy plateau earlier — by approximately Epoch 4 — suggesting that Apple's unified memory architecture facilitates more stable gradient flow in early training phases. CUDA exhibited minor validation loss fluctuations during mid-epochs (3-5), attributed to GPU batch variance and kernel scheduling overhead between the discrete GPU and host CPU.

Classification metrics — Precision (0.96-0.97), Recall (0.96), and F1-Score (0.96) — remained consistent across all backends, and confusion matrices revealed that the primary misclassification pattern occurred between Smoke and Non-Fire classes, reflecting the well-documented visual similarity between smoke and atmospheric haze documented across multiple papers in this review. This consistency validates the hardware-agnostic nature of the ForestFireCNN's feature learning — a finding with significant practical implications for deployment flexibility.

7.5.Analysis of the MPS Performance Advantage

The MPS performance advantage over CUDA in this study requires careful contextualisation. The CUDA benchmark was conducted on an NVIDIA RTX 3050 Notebook GPU — an entry-level mobile discrete

GPU with 4 GB GDDR6 VRAM, subject to thermal and power throttling constraints inherent to notebook implementations. The system was tested in plugged-in mode to maximise available power, while MPS was tested in battery mode — a design choice intended to demonstrate real-world power efficiency. This power mode asymmetry, while deliberate in the study's framing, means that the observed CUDA timing reflects a constrained mobile GPU rather than the class of high-end CUDA hardware (RTX 4090, A100, H100) commonly deployed in research and production deep learning environments.

The MPS advantage is primarily attributable to the unified memory architecture of Apple Silicon: the absence of explicit PCIe data transfer overhead between CPU and GPU memory — a bottleneck that is especially pronounced on notebook discrete GPU implementations — allows the MPS backend to sustain higher effective memory bandwidth to the GPU cores throughout training. For the batch size of 32 and input image resolution of 128x128 used in this study, the dataset transfer overhead represents a meaningful fraction of per-epoch processing time.

8. Comparative Synthesis Across Reviewed Studies

8.1. Model Accuracy and Dataset Context

The accuracy figures reported across the reviewed studies are not directly comparable, as each uses different datasets, class configurations, and evaluation protocols. Nevertheless, a meaningful synthesis can be drawn across the accuracy landscape. Transfer learning with Xception fine-tuning achieved the highest image-level accuracy (98.72%) on a custom 6,911-image forest fire dataset with four classes. The video-based DTA method achieved the highest accuracy (97.92%) on a challenging multi-environment surveillance video benchmark. Compact architectures (ShuffleNetV2-OnFire, ForestFireCNN) achieved 94-96.5% accuracy while operating under significant resource constraints.

8.2. Detection Approach and Temporal Granularity

The reviewed studies collectively span the detection approach spectrum from static single-frame image classification through satellite scene-level detection to long-term video-based temporal accumulation. Static methods offer the simplest deployment path but are most susceptible to false positives from fire-like static objects. Video-based methods with temporal integration demonstrate dramatically reduced false positive rates but require sequential frame access and additional LSTM inference overhead. The optimal approach for any deployment context depends on the trade-off between response latency, false alarm tolerance, available computational resources, and sensor modality.

Table 5: Comparative overview of reviewed studies across key dimensions

Study	Modality	Architecture	Best Accuracy (%)	Key Contribution
Muhammad et al. (2018)	Video (CCTV)	GoogleNet-inspired CNN	94.43	Fine-tuned CNN for surveillance
Kim & Lee (2019)	Video	Faster R-CNN + LSTM	97.92	DTA: temporal fire accumulation
Thomson et al. (2020)	Image/Video	ShuffleNetV2-OnFire	95.0 (97.1 superpixel)	Compact CNN, 155K params, 69fps
Sathishkumar et al. (2023)	Image	Xception + LwF	98.72	Transfer learning + LwF + Bayesian opt.
Zhao et al. (2024)	Satellite image	CNN + InAmp module	94.14 (VIB_SD + InAmp)	Class-specific spectral pattern learning

Kenchgunde et al. (2025)	Image	ForestFireCNN	96.56 (MPS)	MPS vs CUDA vs CPU benchmarking
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8.3. Hardware and Efficiency Considerations

The hardware context of deep learning training and inference for fire detection has been largely unexplored in the literature prior to Kenchgunde et al. (2025). The reviewed application studies — Sathishkumar et al. (2023) used Kaggle GPU-enabled kernels; Thomson et al. (2020) used an NVIDIA 2080Ti; Muhammad et al. (2018) used an NVIDIA GTX TITAN X; Kim and Lee (2019) used an NVIDIA 2080Ti for Faster R-CNN — all relied on conventional CUDA-accelerated training without benchmarking or reporting hardware-specific efficiency metrics. The Kenchgunde et al. (2025) study directly addresses this gap by providing the first controlled cross-platform comparison of MPS, CUDA, and CPU backends for an applied fire detection CNN, establishing that hardware choice primarily affects training speed rather than model quality — a finding with broad implications for the field.

9. Research Gaps and Future Directions

9.1. Identified Research Gaps

Synthesising the reviewed body of literature, several significant research gaps emerge that warrant attention from the research community.

Quantitative energy profiling is entirely absent from all reviewed studies. While Kenchgunde et al. (2025) demonstrate MPS's energy efficiency qualitatively through battery-mode operation, no reviewed study has incorporated hardware power measurement instrumentation to quantify actual watts consumed per unit of training progress. For sustainable AI development and deployment of surveillance systems in power-constrained environments, energy-per-inference and energy-per-training-epoch metrics are critically needed.

Cross-architecture hardware benchmarking remains extremely limited. Kenchgunde et al. (2025) benchmark a single compact CNN; extension to deeper architectures (ResNet50, EfficientNet, Vision Transformers) would reveal whether MPS's advantage scales with model complexity or diminishes as model size exceeds available unified memory capacity.

The intersection of spectral pattern learning and hardware-efficient deployment has not been studied. The InAmp module of Zhao et al. (2024) represents a promising approach to embedding domain knowledge into learnable preprocessing, but its computational overhead on different hardware backends — particularly MPS versus CUDA — has not been characterised.

Multi-generational Apple Silicon benchmarking is absent: Kenchgunde et al. (2025) test only the M4 Pro. Longitudinal comparison across M1, M2, M3, and M4 generations would establish the improvement trajectory of MPS performance for deep learning workloads.

Real-time end-to-end deployment validation — from training to live inference on edge devices — has not been demonstrated for MPS-based systems in a fire detection context. Thomson et al. (2020) demonstrate edge deployment on NVIDIA Xavier-NX but not on Apple Silicon devices.

9.2. Future Research Directions

Based on the identified gaps, the following research directions are recommended.

First, hardware-aware model design for fire detection should be explicitly explored: designing CNN architectures that leverage the specific architectural properties of MPS (unified memory, Neural Engine) to maximise training and inference efficiency for multi-class fire and smoke detection tasks.

Second, integration of spectral preprocessing (InAmp-style) with hardware-optimised inference pipelines on Apple Silicon devices could extend fire smoke detection from satellite imagery to real-time video streams from multi-spectral cameras mounted on UAVs or fixed surveillance platforms.

Third, the LwF continual learning framework explored by Sathishkumar et al. (2023) for domain adaptation in fire detection could be extended to hardware-adaptive deployment: training a model on one hardware configuration and evaluating performance stability when redeploying inference on a different

backend.

Fourth, hybrid spatiotemporal architectures combining compact CNN spatial feature extraction (Thomson et al., 2020) with efficient LSTM temporal integration (Kim and Lee, 2019) should be evaluated specifically on MPS hardware to characterise real-time video-based fire detection performance on portable Apple Silicon devices.

Fifth, large-scale benchmarking studies should extend the approach of Kenchgunde et al. (2025) to include high-end CUDA hardware (RTX 4090, A100, H100) to provide a complete picture of the CUDA versus MPS performance relationship across the full hardware capability range.

10. Conclusion

This review has comprehensively surveyed the landscape of deep learning-based forest fire and smoke detection, integrating findings from six foundational peer-reviewed studies alongside the original cross-platform hardware benchmarking study of Kenchgunde et al. (2025). The central narrative across the reviewed literature is one of progressive capability expansion: from fragile colour-based heuristics and shallow machine learning classifiers to accurate, generalizable deep CNNs; from static single-frame classification to rich spatiotemporal video understanding; from server-dependent CUDA training to on-device Apple Silicon acceleration.

The most consequential finding of our own hardware benchmarking study — that MPS achieves 4x faster training than a notebook CUDA GPU while maintaining statistically identical validation accuracy of approximately 96.5% across all backends — establishes hardware-agnostic model quality as a robust property of well-designed compact CNNs for fire detection. This finding enables practitioners to select hardware based on availability, energy constraints, and cost rather than accuracy considerations, fundamentally democratising access to fire detection AI development.

The complementary contributions of the reviewed studies provide the application and architectural context that situates this hardware finding meaningfully: Sathishkumar et al. (2023) establish that deep CNN accuracy for fire detection can reach 98.72% with transfer learning, LwF, and Bayesian hyperparameter optimisation; Thomson et al. (2020) demonstrate that 95%+ accuracy is achievable with only 155,617 parameters at 40fps on CPU; Kim and Lee (2019) show that temporal accumulation reduces false positive rates to under 3%; Zhao et al. (2024) demonstrate that learned spectral patterns from satellite imagery can substantially improve smoke detection accuracy; and Muhammad et al. (2018) establish the viability of CNN-based surveillance fire detection with real-world false alarm reduction.

Together, these works chart a clear path toward the ultimate goal of this research domain: accurate, real-time, energy-efficient, and widely deployable automated forest fire detection systems capable of providing the early warnings needed to protect ecosystems, communities, and lives from the escalating threat of wildfire.

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